SPE-224715-MS

Streamlining Procurement Through Digital Transformation: Enhancing Efficiency and Compliance

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This paper was prepared for presentation at the GOTECH Conference held in Dubai, UAE, 21 - 23 April, 2025.

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Abstract

OQ Exploration and Production (OQEP) is a leading player in Oman's oil and gas sector, playing a key role in maximizing the value of the nation's energy resources. As part of the OQ Group, OQEP supports Oman's economic growth by optimizing exploration and production efforts. The company's operations are integral to advancing Oman Vision 2040. Currently, OQEP's procurement evaluation process for in-house items relies on multiple manuals, paper-based stages, particularly in the technical bid evaluation. This process is time-consuming, involving extensive data collection and review, leading to delays, technical queries, and potential loss of information. These inefficiencies result in significant time and effort spent on evaluations, with a risk of missing crucial details. To address these challenges, OQEP aims to transition to a more streamlined, digital system. The proposed Artificial Intelligence based system will automate data extraction from bid submissions, evaluate compliance with minimum requirements, and generate reports, significantly improving accuracy and efficiency. It will serve as an open platform for real-time project tracking and issue resolution, accessible to relevant stakeholders. Additionally, the system will act as a comprehensive database for all procurement activities, consolidating historical data for easy management and access. This digital transformation is expected to reduce evaluation time, improve communication between the company and suppliers, and ultimately enhance procurement efficiency, making the process faster, more organized, and more effective.

Introduction

OQ Exploration and Production (OQEP) is a prominent subsidiary of the OQ Group, which plays a vital role in Oman's oil and gas industry. As a leading entity in the exploration and production sector, OQEP is dedicated to maximizing the value of the country's energy resources, contributing significantly to Oman's economic development. The company is engaged in the full spectrum of upstream activities, including the exploration, drilling, and production of oil and gas. OQEP's operations are integral to the broader strategic goals outlined in Oman Vision 2040, which aims to diversify the nation's economy and increase the value of its energy exports. By utilizing advanced technologies and optimizing its operations, OQEP helps improve the efficiency and sustainability of Oman's energy resources. The company is also committed to environmental responsibility and has implemented several initiatives to reduce its ecological footprint.

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With its strong focus on innovation and operational excellence, OQEP plays a critical role in shaping the future of Oman's energy sector while supporting the nation's long-term growth and development.

A critical aspect of OQEP's projects involves its procurement evaluation process, which encompasses the sourcing and evaluation of items and services required for its energy projects. Despite the importance of procurement, the current system is largely manual and paper-based, making it prone to inefficiencies, delays, and human errors. The technical bid evaluation process, in particular, is labor-intensive, requiring the review of detailed submissions, data collection, and compliance verification.

The time and effort spent on manual method to perform the technical bid evaluation in the oil and gas industry significantly hamper efficiency, productivity, and employee satisfaction. With manual processes contributing to 60%-70% longer evaluation times, increased risk of human errors which lead to cost implications. The technical evaluation process challenges are summarized as follows:

- **Time Consumption in Manual Evaluation**

  I. Average Time Spent on Bid Template Preparation:
  The preparation of technical bid evaluation templates is a time-intensive process, requiring approximately 5 to 15 hours depending on the complexity of items to be evaluated. This process involves gathering detailed specifications, organizing data into a structured format, and ensuring consistency and accuracy across all templates. Additionally, coordination between multiple departments and stakeholders for input further extends the preparation time.

  II. Average Time Spent on Bid Evaluation:
  Manual bid evaluation can take up to 60%-70% longer than expected automated methods. Reviewing each bid manually can take anywhere from 8 to 10 hours per bid. Paper-based processes are significantly slower.

  III. Time Wasted on Compliance Checks:
  The manual data entry, evaluation, and verification can add 30%-40% more time to the technical evaluation. Since oil and gas bids typically involve a large volume of technical data, verifying compliance with technical requirements manually can consume substantial hours.

  IV. Average time spent on Technical Queries preparation:
  The manual preparation of technical queries significantly extends the evaluation process, adding an additional 10%-15% to the overall timeline. This includes drafting the queries, communicating them to vendors, and subsequently receiving, reviewing, and analyzing vendor responses. The lack of automation increases the time and effort required, often leading to inefficiencies and delays.

  V. Average time Spent to complete the Evaluation:
  Based on the processes outlined in points I through IV, the company typically experiences a delay of 4-5 weeks in completing the bid evaluation process. This timeframe excludes additional delays caused by vendor response times to technical queries, which further prolongs the overall timeline and impacts project schedules.

- **Employee Effort and Efficiency**

  I. High Employee Workload:
  Technical team in the oil and gas industry can spend 70%-80% of their time on manual data entry, review, and bid evaluation, which significantly detracts from their ability to focus on other assigned tasks. In complex item procurements, the number of employees involved in manual bid evaluations can be extensive, typically, include 4-5 employees per item procurement.

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II. Manual Errors and Rework:
Manual evaluations often lead to errors in data entry or compliance checks, which must be corrected through rework. The time spent on rework can double the effort required for initial evaluations.

III. Risk of Missing Critical Information:
Through the manual method, there is an increased risk of missing critical data in bid submissions, which can lead to delays. Manual processes in procurement evaluation can cause up to a 20%-25% loss of critical information across multiple stages of evaluation.

- **Unpredicted Cost**
  Inefficiencies, delays, and human errors in manual procurement evaluation processes can result unpredicted increase in procurement costs, primarily due to design changes after the award stage, which lead to cost implications.
- **Inefficiencies in Communication**
  Manual processes involve continuous back-and-forth communication between technical team, project team, contract team, suppliers, and OEM, leading to delays from slow responses to queries. Additionally, the transfer of queries and special requests to bidders can be missed due to the manual communication through emails.

These challenges highlight the need for a more efficient and modern approach to perform the technical evaluation process, leading OQEP to seek a digital solution to address these challenges.

This paper presents the rationale for digitalizing OQEP's procurement processes, with a focus on the automation of technical bid evaluations. It explores the current challenges, the proposed digitalization (e.g. AI-based solution), and the expected improvements in efficiency, accuracy, and communication throughout the procurement lifecycle.

Statement of Theory and Definitions

The primary theory underlying the digital transformation at OQEP is the concept of Automation and Artificial Intelligence applied to business processes. AI refers to the use of machine learning algorithms to perform tasks that typically require human intelligence, such as data analysis and decision-making. In procurement evaluation process, AI can be used to automate tasks like data extraction, data entry, compliance checks, technical queries preparation and final report generation, improving speed and accuracy. Key terms in this paper include:

- **Technical Bid Evaluation:** The process of assessing bids from suppliers based on predefined criteria, technical data and specifications.
- **Artificial Intelligence (AI):** Machine learning algorithms used to automate processes such as data extraction and compliance evaluation.
- **Automation:** The use of technology to perform tasks that traditionally require human intervention.

Description and Application of Digitalize the Process

The AI-based digital system proposed by OQEP is designed to revolutionize the company's procurement evaluation process by automating the data extraction and compliance evaluation stages, which will significantly enhance operational efficiency, accuracy, and speed. This digitalized system uses advanced machine learning algorithms that are tailored to the specific needs of the oil and gas industry, optimizing the procurement evaluation lifecycle.

System Overview. The digitalized system will enable smooth data flow between the various stages of the procurement evaluation process. It begins by automatically extracting predefined data from technical

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bid submissions, including technical specifications, vendor names and key compliance documents. The system is designed to verify that each bidder meets the minimum compliance requirements. These requirements typically include documents such as a Table of Compliance for technical specifications, Quality Requirements, Health, Safety, and Environmental (HSE) standards, and In-Country Value (ICV) details. By automating the extraction, entering and verification of these key pieces of information, the system reduces the likelihood of human error and significantly speeds up the process.

Stage 1: Go-No-Go Stage (Initial Compliance Filtering). In the initial stage of the technical evaluation, referred to as the "Go-No-Go" stage, the system filters bidders based on a predefined set of compliance criteria. The system evaluates whether each bidder has submitted all necessary documents and whether the data provided meets the basic requirements for the project. This stage serves as a gatekeeping mechanism, ensuring that only bidders who comply with the company initial bids submissions documentation move forward in the technical evaluation process.

The system's algorithms cross-check the submitted documents against the compliance criteria, flagging any missing or incomplete documents, such as Signed and stamped Table of Compliance for technical specifications, Reference Experience List for supplying the requested material, HSE requirements, ICV plan and quality certifications. For example, if a bidder fails to meet specific HSE requirements or does not submit an adequate ICV plan, the system will automatically reject their bid and exclude them from the subsequent evaluation stages. This reduces the time spent by technical teams manually reviewing submissions and helps prevent unnecessary delays.

At the conclusion of this stage, the system generates a list of qualified bidders who have met all the initial requirements and are eligible to move on to the next phase of evaluation. This list is essential for focusing the resources of the technical evaluation team on the most qualified bidders.

Stage 2: Detailed Technical Evaluation. Once the qualified bidders are identified, the system moves into the second stage, which is the detailed technical evaluation. At this point, the system compares the detailed technical offers submitted by the bidders against a set of predetermined, project-specific technical requirements. These requirements typically include detailed specifications, such as design parameters, equipment standards, performance metrics, and adherence to industry best practices.

Using machine learning, the system can quickly identify compliance or deviations from these technical requirements. It generates detailed compliance reports that show where each bidder’s submission aligns or diverges from the specified technical criteria. For instance, if a bidder’s technical submission differs in terms of material quality, testing requirements, or design specifications, the system will highlight these discrepancies and provide a clear comparison for each bidder’s proposal. This transparency enables the evaluation team to easily assess how well each bidder meets the required technical specifications.

Additionally, the system allows users to generate technical queries for each bidder. For example, if a bidder’s proposal contains discrepancies, the system will create a technical queries report that can be used to send to the bidder for clarification or further explanation, these queries can be modified, detailed and evaluated by the focal technical person. This enables the technical team to have organized list of technical queries, ensuring that all queries are tracked, responded to, and resolved. This process eliminates the need of manual listing of technical queries and reduces the risk of missing critical information.

After all queries are addressed and the necessary clarifications are made, the system generates a finalized technical evaluation report and issue it to be electronically singed by all evaluation stakeholders. This report includes a list of disqualified bidders and technically qualified bidders who meet all the necessary specifications and are eligible to proceed to the commercial evaluation phase.

Collaboration and Real-Time Input from the Technical Team. One of the key benefits of the proposed AI-based system is the ability to facilitate collaboration among multiple stakeholders involved in the technical evaluation process. The system will allow all members of the technical evaluation team to access the

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same evaluation document in real time, ensuring that each team member can simultaneously input their observations, evaluations, and technical queries into the system.

This collaborative feature promotes greater transparency and coordination among the evaluation team. Team members can view each other's inputs, add their own comments, and raise concerns directly within the shared evaluation sheet. By enabling real-time updates, the system minimizes the risk of mismatched information or discrepancies between team members' assessments. This feature also ensures that everyone involved in the evaluation is on the same page, which is crucial when dealing with complex technical specifications and large amounts of data.

Furthermore, the system's collaborative approach ensures that any discrepancies or deviations from the specifications are flagged and addressed by the relevant experts. For example, if an issue arises regarding the interpretation of a particular requirement, the system allows technical experts from different disciplines to communicate and come to an agreement, thus reducing potential misunderstandings and enhancing the overall quality of the evaluation.

Management follow-up and Timeline Tracking. The system also offers significant advantages for management follow-up and project tracking. The digital platform provides a comprehensive overview of the procurement process, showing the current status and the stage of completion for each bid evaluation. A timeline is displayed, outlining key milestones and deadlines for each stage of the evaluation process. This transparency enables managers to easily monitor the progress of the technical evaluation process and ensure that it stays on track.

Moreover, the system includes a built-in review cycle for technical queries, ensuring that each query undergoes a set period of review before being closed. The review cycle timing is clearly displayed, allowing management to track how long each query takes to resolve and whether the technical evaluation process is being completed within the expected timeframe. This feature provides an added layer of accountability, helping to prevent delays and ensuring that the technical evaluation process meets deadlines.

Centralized Database. The AI-based digital system will function as a centralized database for all procurement activities at OQEP, ensuring that every aspect of the technical evaluation process is thoroughly documented and easily accessible. This database will consolidate all technical evaluation data, including bid submissions, compliance reports, technical evaluations, communications, and historical records, into a single, organized platform.

By centralizing this information, the system will provide stakeholders with an access to critical data, improving transparency and decision-making. The system will store comprehensive records of all technical evaluation process, which can be referenced for future projects, audits, or performance reviews. Additionally, this centralized approach will streamline data management, allowing for quick retrieval of information, reducing the risk of lost or misplaced documents, and ensuring that all procurement activities are accurately tracked and maintained for long-term operational efficiency.

Presentation of Data and Results

The primary data generated from the implementation of the AI-based system will focus on key performance indicators, such as the speed and accuracy of technical bid evaluations, when compared to the current manual process. The expected results include the following:

- **Reduction in Evaluation Time**: One of the most significant improvements expected from the AI system is the reduction in the time spent on bid evaluations. Currently, manual evaluations require extensive effort to extract data, cross-check compliance, and review each submission individually. With AI-powered automation, data extraction and compliance checking can be completed in a fraction of the time. The system will automatically analyze and categorize bid data, flagging any discrepancies, which enables technical evaluation teams to focus on making informed decisions

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rather than spending valuable time on administrative tasks. As a result, completing the evaluation will be faster, and procurement cycles will shorten, allowing OQEP to handle more procured items in less time, driving overall efficiency.

- **Improved Accuracy**: The system is designed to minimize human error by automating the data entry and compliance verification processes. In the current manual process, there is a risk of overlooking key details or making mistakes while transcribing and evaluating complex bid information. By removing these manual steps, the AI system ensures greater consistency and reliability in compliance evaluations, leading to more accurate results. This will help avoid costly errors that may arise from misunderstandings or missing important details, ensuring that only compliant bids proceed to the next stages of the technical evaluation process.
- **Enhanced Communication**: Effective communication is critical to a successful technical evaluation process, but manual systems often lead to delays and misunderstandings due to the reliance on emails, phone calls, and fragmented communication channels. The AI system, however, provides a centralized digital platform for real-time tracking. Project managers, procurement teams, and other stakeholders will have access to up-to-date information on the status of bid evaluations, including technical queries, clarifications, and updates from bidders. This transparency will significantly reduce delays caused by miscommunication, ensuring that all parties are aligned and informed throughout the process. Additionally, the platform will support instant notifications and reminders, ensuring that technical teams respond quickly to any issues or queries, further reducing response time.
- **Document and Report Generation**: The system will automate the creation of essential procurement documents, including technical evaluation reports, compliance summaries, and other related documentation. In the manual process, generating these reports often involves extensive copying, pasting, and compiling of data from multiple sources, which can be time-consuming and prone to inconsistencies. The AI system will streamline this process by automatically compiling and formatting reports based on the data entered into the system, ensuring consistency across all documentation. This automated document generation will not only save significant time and effort but will also improve the overall professionalism and accuracy of the reports. Furthermore, these automated reports will be stored in the centralized database, allowing for easy access and retrieval for audits, historical reference, or future procurement projects.
- **Pilot phase**

  The AI based solution shall be tested through a pilot phase before full implementation to ensure the functionality and accuracy of the system to avoid systematic errors which can lead to fail the objective of the digitized solution. The Test Procedures for the AI-based solution will include all the defined stages and expected results.

By addressing these areas of improvement—evaluation time, accuracy, communication, and documentation—the AI-based system will lead to a more efficient, effective, and organized procurement process at OQEP, ultimately contributing to better decision-making, cost savings, and smoother project execution.

Conclusions

The digital transformation of procurement at OQEP, driven by an Artificial Intelligence system, represents a significant step toward improving efficiency, accuracy, and communication in the Company's projects. By automating technical bid evaluations, the Company can reduce manual effort, accelerate procurement cycles, and enhance the decision-making process. The results of this transformation are expected to lead to a more efficient procurement process, improving collaboration among stakeholders and ensuring that critical information is not lost between stages. This move toward digitalization is aligned with OQEP's

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goals of operational excellence and resource optimization, ultimately contributing to the company's long-term competitiveness and supporting Oman's Vision 2040.

The successful implementation of this AI-driven system will demonstrate how technology can be leveraged to streamline complex procurement processes and enhance overall business operations. Further research and testing will be necessary to refine the system, but the early results indicate that this digital solution has the potential to deliver substantial improvements to OQEP's technical evaluation process and beyond.

This paper provides an in-depth exploration of the digitalization of procurement at OQEP, emphasizing the role of AI in transforming traditional workflows into more efficient, data-driven processes. The results of this transformation will not only benefit OQEP but could also have significant implications for other organizations in the oil and gas sector looking to modernize their procurement practices.

Acknowledgment

We would like to express our gratitude to OQ Exploration & Production (OQEP) for their support and resources, which were instrumental in the completion of this work. Special thanks to the management and technical teams for their valuable insights and encouragement throughout this project.

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